One way ANOVA

Fixed Effects Model and ANOVA

Table 13.2 Typical data for a single-factor experiment

Treatment	*Observation (response)*						*Total*	*Average*
1	y_{11}	y_{12}	$\cdots$	y_{1j}	$\cdots$	y_{1n}	$y_{1.}$	$\bar{y}_{1.}$
2	y_{21}	y_{22}	$\cdots$	y_{2j}	$\cdots$	y_{2n}	$y_{2.}$	$\bar{y}_{2.}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$	$\vdots$
i	y_{i1}	y_{i2}	$\cdots$	y_{ij}	$\cdots$	y_{in}	$y_{i.}$	$\bar{y}_{i.}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$	$\vdots$	$\vdots$
k	y_{k1}	y_{k2}	$\cdots$	y_{kj}	$\cdots$	y_{kn}	$y_{k.}$	$\bar{y}_{k.}$
			$\cdots$				$y_{..}$	$\bar{y}_{..}$

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}; \qquad i = 1, 2, \ldots, k \quad \text{and} \quad j = 1, 2, \ldots, n$$

Fixed Effects Model and ANOVA

If k treatments are chosen specifically, we can test the hypothesis only about these treatment means and conclusions cannot be extended to similar treatments that were not considered. The model under this situation is called the *fixed-effects model.*

In this model, the treatment effects τ_i are usually defined as deviations from overall mean, and hence we have $\sum_{i=1}^{k} \tau_i = 0$. Therefore, in this case the null hypothesis and alternative hypothesis become

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k = \mu \Rightarrow H_0 : \tau_1 = \tau_2 = \cdots = \tau_k = 0$$

$$H_1 : \text{At least two means are not equal} \Rightarrow H_1 : \tau_i \neq 0 \text{ for at least one } i.$$

Table 13.3 ANOVA for fixed effects model

Source of variation	*SS*	*df*	*MSS*	*E(MSS)*	*F-ratio*
Between treatments	SSTr	$k-1$	$\text{MSSTr} = \dfrac{\text{SSTr}}{k-1}$	$E(\text{MSSTr}) = \sigma^2 + \dfrac{n\sum_{i=1}^{k}\tau_i^2}{k-1}$	$F_c = \dfrac{\text{MSSTr}}{\text{MSSE}}$
Within treatments (error)	SSE	$k(n-1)$	$\text{MSSE} = \dfrac{\text{SSE}}{k(n-1)}$	$E(\text{MSSE}) = \sigma^2$	
Total	TSS	$kn-1$			

$$\text{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{kn}$$

$$\text{SSTr} = \frac{\sum_{i=1}^{k} y_{i.}^2}{n} - \frac{y_{..}^2}{kn}$$

$$\text{SSE} = \text{TSS} - \text{SSTr} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{\sum_{i=1}^{k} y_{i.}}{n}$$

Fixed Effects Model and ANOVA

Note: (*Unbalanced Case*) If the number of observations n is not equal for all the treatments, then we may have $n_1, n_2, \cdots, n_i, \cdots, n_k$ as the number of observations for $i = 1, 2, \ldots, k$ treatments, respectively. And now we will have each observation as Y_{ij}, $i = 1, 2, \ldots, k$ and $j = 1, 2, \ldots, n_i$. Let $N = \sum_{i=1}^{k} n_i$, then the sums of squares can be obtained as follows:

$$\mathrm{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \frac{y_{..}^2}{N}, \quad \text{where } y_{..} = \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij} \tag{13.26}$$

$$\mathrm{SSTr} = \sum_{i=1}^{k} \frac{y_{i.}^2}{n_i} - \frac{y_{..}^2}{N}, \quad \text{where } y_{i.} = \sum_{j=1}^{n_i} y_{ij}, \quad i = 1, 2, \ldots, k \tag{13.27}$$

$$\mathrm{SSE} = \mathrm{TSS} - \mathrm{SSTr} = \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \sum_{i=1}^{k} \frac{y_{i..}^2}{n_i} \tag{13.28}$$

Random Effects Model and ANOVA

In random effects model, the population, being large, testing hypothesis about individual treatment effects is meaningless, so instead we test the hypothesis about the variance component σ_τ^2. That is,

$$H_0 = \sigma_\tau^2 = 0 \quad \text{versus} \quad H_1 = \sigma_\tau^2 > 0$$

If $\sigma_\tau^2 = 0$, then all treatments are identical, but if $\sigma_\tau^2 > 0$ variability exists between treatments.

Table 13.4 ANOVA for random effects model

Source of variation	*SS*	*df*	*MSS*	*E(MSS)*	*F-ratio*
Between treatments	SSTr	$k-1$	$\text{MSSTr} = \dfrac{\text{SSTr}}{k-1}$	$E(\text{MSSTr}) = \sigma^2 + n\sigma_\tau^2$	$F_c = \dfrac{\text{MSSTr}}{\text{MSSE}}$
Within treatments (error)	SSE	$k(n-1)$	$\text{MSSE} = \dfrac{\text{SSE}}{k(n-1)}$	$E(\text{MSSE}) = \sigma^2$	
Total	TSS	$kn-1$			

The following table gives the breaking strength (in kilogram) of 100 leas, 20 each from five machines available in a spinning mill. Perform the analysis of variance to test whether the yarn produced on these machines differ in breaking strength from one other. Test the hypothesis at both 5% and 1% levels of significance. If the null hypothesis is rejected, then perform the multiple comparison test.

	Machine				
Observation	*A*	*B*	*C*	*D*	*E*
1	23	26	34	26	36
2	30	33	36	30	25
3	21	22	32	25	38
4	24	31	24	33	24
5	31	29	30	24	35
6	24	33	29	27	26
7	25	29	25	32	31
8	25	33	25	29	25
9	28	26	29	24	37
10	25	25	32	25	28
11	31	34	27	31	38
12	30	31	32	32	25
13	28	36	31	23	36
14	23	32	33	28	30
15	29	23	24	26	32
16	24	35	27	30	36
17	30	35	33	26	25
18	22	25	30	32	33
19	25	37	27	23	29
20	23	26	31	30	23
Total	521	601	591	556	612
Mean	26.00	30.05	29.60	27.80	31.00

Solution

Since there are only five machines available, our conclusion will be pertaining to only the specific five machines. ANOVA under the fixed effects model is used.

Null and Alternative Hypotheses

From the data, it may be noted that there are five ($k = 5$) treatments, each with twenty randomly selected observations ($n = 20$). Therefore, we have

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu \Rightarrow H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = \tau_5 = 0$$

That is, under the null hypothesis, we say that there is no significant difference between the machines with regard to breaking strength.

If null hypothesis is rejected, then we say that there is a significant difference between the machines with regard to breaking strength.

$$H_1\text{: At least two means are not equal} \Rightarrow H_1: \tau_i \neq 0 \quad \text{for at least one } i.$$

Computation of Sums of Squares

Now, using the Eqs. (13.23)–(13.25), the sums of squares can be computed as follows:

(i) Total sum of squares is

$$\text{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{kn}$$

where

$$\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 = 23^2 + 30^2 + 21^2 + \cdots + 29^2 + 23^2 = 84837$$

and

$$\frac{y_{..}^2}{kn} = \frac{1}{kn}\left(\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}\right)^2 = \frac{1}{(5)(20)}(23+30+\cdots+29+23)^2 = \frac{(2881)^2}{100} = 83001.61$$

Therefore,

$$\text{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{\text{kn}} = 84837 - 83001.61 = 1835.39$$

(ii) Treatment sum of squares is

$$\text{SSTr} = \frac{\sum_{i=1}^{k} y_{i.}^2}{n} - \frac{y_{..}^2}{kn}$$

where

$$\frac{\sum_{i=1}^{k} y_{i.}^2}{n} = \frac{1}{n}(y_{1.}^2 + y_{2.}^2 + \cdots + y_{k.}^2) = \frac{1}{n}\left[\left(\sum_{j=1}^{n} y_{1j}\right)^2 + \left(\sum_{j=1}^{n} y_{2j}\right)^2 + \cdots + \left(\sum_{j=1}^{n} y_{kj}\right)^2\right]$$

Here, $n = 20$ and $k = 5$. Therefore, we have

$$\frac{\sum_{i=1}^{k} y_{i.}^2}{n} = \frac{1}{20}(521^2 + 601^2 + 591^2 + 556^2 + 612^2) = 83280.15$$

Therefore,

$$\text{SSTr} = 83280.15 - 83001.61 = 278.54$$

(iii) Error sum of square is

$$\begin{aligned} ESS &= TSS - SSTr \\ &= \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{\sum_{i=1}^{k} y_{i.}^2}{n} \\ &= 1835.39 - 278.54 \\ &= 84837 - 83280.15 \\ &= 1556.85 \end{aligned}$$

Degrees of Freedom (df)

For total: $nk - 1 = 100 - 1 = 99$

For treatment: $k - 1 = 5 - 1 = 4$

For error: df (Total) – df (Treatment) = 99 – 4 = 5

Construction of ANOVA Table

Now, the following ANOVA table is constructed based on the above computations and the degrees of freedom.

Source of variation	*SS*	*df*	*MSS*	*F-ratio*
Between machines (treatments)	278.54	4	69.64	$F_c = \frac{69.64}{16.39} = 4.25$
Within machines (error)	1556.85	95	16.39	
Total	1835.39	99		

F table

	$\alpha = 0.05$																		
	d.f.N.																		
d.f.D.	***1***	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***12***	***15***	***20***	***24***	***30***	***40***	***60***	***120***	**∞**
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84

Example 13.2

The following table shows the result of a sock length felting shrinkage experiment, the figures being percentage shrinkage after treatment in a wash wheel. The wash wheel takes four socks at a time and so the results are divided into six groups of four, corresponding to the six different runs on the machine. There are six runs so that twenty-four socks are tested in all. The division of socks into groups of four was done at random. Perform ANOVA. If hypothesis is rejected, then perform the multiple range test.

	Run number					
Socks	*1*	*2*	*3*	*4*	*5*	*6*
1	9.5	4.3	6.5	6.1	10.0	9.3
2	8.8	7.8	8.3	7.3	4.8	8.7
3	11.4	3.2	8.6	4.2	5.4	7.2
4	7.8	6.5	8.2	4.1	9.6	10.1
Total	37.5	21.8	31.6	21.7	29.8	35.3

It may be noted from this example that, the six runs are chosen randomly from the population of infinite number of runs. Also, the socks per group are assigned randomly to each run. Therefore, whatever the conclusions we draw from the analysis are appropriate for the entire population of runs, and hence ANOVA under the random effects model is used. For random effects model, the null hypothesis is that the variance due to treatment effects σ_τ^2 becomes zero. That is,

$$H_0:\sigma_\tau^2=0$$

This implies that there is no significant difference between the runs with regard to shrinkage of length after wash. Or otherwise, we assume that a particular run will not influence significantly the shrinkage in length. The alternative hypothesis becomes

$$H_1:\sigma_\tau^2>0$$

That is, if the null hypothesis is rejected then we say that there is a significant difference between the machines with regard to breaking strength.

(i) Total sum of squares is

$$\text{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{kn}$$

where

$$\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 = 9.5^2 + 4.3^2 + 21^2 + \cdots + 9.6^2 + 10.1^2 = 1427.99$$

and

$$\frac{y_{..}^2}{kn} = \frac{1}{kn}\left(\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}\right)^2 = \frac{1}{(6)(4)}(9.5 + 4.3 + \cdots + 9.6 + 10.1)^2 = \frac{(177.7)^2}{24} = 1315.72$$

Therefore,

$$\text{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{kn} = 1427.99 - 1315.72 = 112.27$$

Treatment sum of squares is

$$\mathrm{SSTr} = \frac{\sum_{i=1}^{k} y_{i.}^2}{n} - \frac{y_{..}^2}{kn}$$

where

$$\frac{\sum_{i=1}^{k} y_{i.}^2}{n} = \frac{1}{n}(y_{1.}^2 + y_{2.}^2 + \cdots + y_{k.}^2) = \frac{1}{n}\left[\left(\sum_{j=1}^{n} y_{1j}\right)^2 + \left(\sum_{j=1}^{n} y_{2j}\right)^2 + \cdots + \left(\sum_{j=1}^{n} y_{kj}\right)^2\right]$$

Here, $n = 4$ and $k = 6$. Therefore, we have

$$\frac{\sum_{i=1}^{k} y_{i.}^2}{n} = \frac{1}{4}(37.5^2 + 21.8^2 + 31.6^2 + 21.7^2 + 29.8^2 + 35.3^2) = 1371.27$$

Therefore,

$$\mathrm{SSTr} = 1371.27 - 1315.72 = 55.55$$

(iii) Error sum of square is

$$ESS = TSS - SSTr$$

$$= \sum_{i=1}^{k} \sum_{j=1}^{n} y_{ij}^2 - \frac{\sum_{i=1}^{k} y_{i.}^2}{n}$$

$$= 112.27 - 55.55$$

$$=1427.99 - 1371.27$$

$$= 56.72$$

Degrees of Freedom (df)

For total: $nk - 1 = 24 - 1 = 23$

For treatment: $k - 1 = 6 - 1 = 5$

For error: df (Total) – df (Treatment) = $23 - 5 = 18$

Construction of ANOVA Table

Now, the following ANOVA table is constructed based on the above computations and the degrees of freedom.

Source of variation	*SS*	*df*	*MSS*	*F-ratio*
Between runs (treatments)	55.55	5	11.11	$F_c = \frac{11.11}{3.15} = 3.53$
Within runs (error)	56.72	18	3.15	
Total	112.27	23		

Inference

Since the calculated F-value, $F_c = 3.53$, is greater than the tabulated value at 5% $[F_t = F(0.05; 5, 18) = 2.773]$ (refer to Figure 13.4 and Table E.2 in Appendix E) the hypothesis is rejected. This implies that $\sigma_\tau^2 > 0$ and hence the influence of the runs is significantly different.

Two way ANOVA

Fixed Effects Model for Two way ANOVA

Table 13.7 The rearranged data for a CRBD experiment

Treatment	*1*	*2*	⋯	*Block$_j$*	⋯	*n*	*Total*	*Mean*
1	y_{11}	y_{12}	⋯	y_{1j}	⋯	y_{1n}	$y_{1.}$	$\bar{y}_{1.}$
2	y_{21}	y_{22}	⋯	y_{2j}	⋯	y_{2n}	$y_{2.}$	$\bar{y}_{2.}$
⋮	⋮	⋮		⋮		⋮	⋮	⋮
i	y_{i1}	y_{i2}	⋯	y_{ij}	⋯	y_{in}	$y_{i.}$	$\bar{y}_{i.}$
⋮	⋮	⋮		⋮		⋮	⋮	⋮
k	y_{k1}	y_{k2}	⋯	y_{kj}	⋯	y_{kn}	$y_{k.}$	$\bar{y}_{k.}$
Total	$y_{.1}$	$y_{.2}$	⋯	$y_{.j}$	⋯	$y_{.n}$	$y_{..}$	
Mean	$\bar{y}_{.1}$	$\bar{y}_{.2}$	⋯	$\bar{y}_{.j}$	⋯	$\bar{y}_{.n}$		$\bar{y}_{..}$

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}; \quad i = 1, 2, \ldots, k \quad \text{and} \quad j = 1, 2, \ldots, n$$

Fixed Effects Model for Two way ANOVA

In this model, the treatment effects τ_i are usually defined as deviations from overall mean, and hence we have $\sum_{i=1}^{k} \tau_i = 0$ which implies

$$\mu_{i.} = \frac{1}{n}\sum_{j=1}^{n}(\mu + \tau_i + \beta_j) = \mu + \tau_i \tag{13.35}$$

Therefore, in this case the null hypothesis and alternative hypothesis become

$H_{01}: \mu_{1.} = \mu_{2.} = \cdots = \mu_{k.} = \mu \Rightarrow H_{01}: \tau_1 = \tau_2 = \cdots = \tau_k = 0$

H_{11}: At least two treatment means are not equal $\Rightarrow H_{11}: \tau_i \neq 0$ for at least one i.

Fixed Effects Model for Two way ANOVA

Table 13.8 ANOVA for fixed effects model of CRBD

Source of variation	*SS*	*df*	*MSS*	*E(MSS)*	*F-ratio*
Between treatments	SSTr	$k-1$	$\text{MSSTr} = \dfrac{\text{SSTr}}{k-1}$	$E(\text{MSSTr}) = \sigma^2 + \dfrac{n\sum_{i=1}^{k}\tau_i^2}{k-1}$	$F_{c1} = \dfrac{\text{MSSTr}}{\text{MSSE}}$
Between Blocks	SSB	$n-1$	$\text{MSSB} = \dfrac{\text{SSB}}{k-1}$	$E(\text{MSSB}) = \sigma^2 + \dfrac{k\sum_{j=1}^{n}\beta_j^2}{n-1}$	$F_{c2} = \dfrac{\text{MSSB}}{\text{MSSE}}$
Error	SSE	$(k-1)(n-1)$	$\text{MSSE} = \dfrac{\text{SSE}}{k(n-1)}$	$E(\text{MSSE}) = \sigma^2$	
Total	TSS	$kn-1$			

Computation of Sum of Squares

$$TSS = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{kn}$$

$$SSB = \frac{\sum_{j=1}^{n} y_{.j}^2}{k} - \frac{y_{..}^2}{kn}$$

$$SSTr = \frac{\sum_{i=1}^{k} y_{i.}^2}{n} - \frac{y_{..}^2}{kn}$$

$$SSE = TSS - SSTr - SSB = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{\sum_{i=1}^{k} y_{i.}}{n} - \frac{\sum_{j=1}^{n} y_{.j}}{k} + \frac{y_{..}^2}{kn}$$

Random Effects Model for Two way ANOVA

In the random effects model, we know that ε_{ij} are assumed to be normally independently distributed with mean 0 and variance σ^2, the treatment effects τ_i are normally independently distributed with mean 0 and variance σ_τ^2 and block effects β_j are normally distributed with mean 0 and variance σ_β^2. It may be noted that ε_{ij} are independent from both τ_i and β_j that are independent. Also, it is known that $E(\tau_i)=0 \Rightarrow \sigma_\tau^2 = E(\tau_i^2)$, $E(\sum_{i=1}^{k} \tau_i^2) = k\sigma_\tau^2$, $E(\beta_j)=0$ $\Rightarrow \sigma_\beta^2 = E(\beta_j^2)$, $E(\sum_{j=1}^{n} \beta_j^2) = n\sigma_\beta^2$, $E(\varepsilon_{ij})=0 \Rightarrow \sigma^2 = E(\varepsilon_{ij}^2)$, $E(\varepsilon_{i.}^2) = n\sigma^2$ and $E(\varepsilon_{..}^2) = kn\sigma^2$.

Therefore the null hypotheses and alternative hypotheses are given as

(i) For treatment effects: $H_{01} = \sigma_\tau^2 = 0$ versus $H_{11} = \sigma_\tau^2 > 0$

(ii) For block effects: $H_{02} = \sigma_\beta^2 = 0$ versus $H_{12} = \sigma_\beta^2 > 0$

Random Effects Model for Two way ANOVA

Table 13.9 ANOVA for random effects model of CRBD

Source of variation	*SS*	*df*	*MSS*	*E(MSS)*	*F-ratio*
Between treatments	SSTr	$k-1$	$\text{MSSTr} = \dfrac{\text{SSTr}}{k-1}$	$E(\text{MSSTr}) = \sigma^2 + n\sigma_\tau^2$	$F_{c1} = \dfrac{\text{MSSTr}}{\text{MSSE}}$
Between blocks	SSB	$n-1$	$\text{MSSB} = \dfrac{\text{SSB}}{k-1}$	$E(\text{MSSB}) = \sigma^2 + k\sigma_\beta^2$	$F_{c2} = \dfrac{\text{MSSB}}{\text{MSSE}}$
Error	SSE	$(k-1)(n-1)$	$\text{MSSE} = \dfrac{\text{SSE}}{k(n-1)}$	$E(\text{MSSE}) = \sigma^2$	
Total	TSS	$kn-1$			

In order to know the effect of four randomly chosen brands of fertilizers on the yield of potato, a piece of field is chosen and it is divided into four column and five rows. To control the variability arising from a nuisance factor (block), all the four fertilizers are tried once in each block. The fertilizers are, in fact, randomly used in blocks. However, the treatment-wise data on yield (in kilograms) are recorded as shown in the following table. Perform an analysis of variance to test the significance of the difference between the four brands of fertilizers with regard to yield.

	Fertilizer				
Block	***A***	***B***	***C***	***D***	***Total***
1	50	35	45	37	167
2	45	38	43	35	161
3	48	41	40	39	168
4	47	40	39	40	166
5	42	38	37	40	157
Total	232	192	204	191	819

Null and Alternative Hypotheses

From the above table, it may be noted that there are four ($k = 4$) treatments and five ($n = 5$) blocks. Therefore, we have

$$H_{01}: \mu_{1.} = \mu_{2.} = \mu_{3.} = \mu_{4.} = \mu \Rightarrow H_{01}: \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$$

H_{11}: At least two treatment means are not equal $\Rightarrow H_{11}: \tau_i \neq 0$ for at least one i

That is, under the null hypothesis, we say that there is no significant difference between the treatments with regard to yield.

$$H_{02}: \mu_{.1} = \mu_{.2} = \mu_{.3} = \mu_{.4} = \mu_{.5} = \mu \Rightarrow H_{02}: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

H_{12}: At least two block means are not equal $\Rightarrow H_{12}: \beta_j \neq 0$ for at least one j

Under this null hypothesis, we say that there is no significant difference between the blocks with regard to yield.

(i) Total sum of squares is

$$\text{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{kn}$$

where

$$\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 = 50^2 + 45^2 + 18.4^2 + \cdots + 40^2 + 40^2 = 33875$$

and

$$\frac{y_{..}^2}{kn} = \frac{1}{kn}\left(\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}\right)^2 = \frac{1}{(4)(5)}(50 + 45 + \cdots + 40 + 40)^2 = \frac{(819)^2}{20} = 33538.05$$

Therefore,

$$\text{TSS} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{..}^2}{kn} = 33875 - 33538.05 = 336.95$$

Treatment sum of squares is

$$\mathrm{SSTr} = \frac{\sum_{i=1}^{k} y_{i.}^2}{n} - \frac{y_{..}^2}{kn}$$

where

$$\frac{\sum_{i=1}^{k} y_{i.}^2}{n} = \frac{1}{n}(y_{1.}^2 + y_{2.}^2 + \cdots + y_{k.}^2) = \frac{1}{n}\left[\left(\sum_{j=1}^{n} y_{1j}\right)^2 + \left(\sum_{j=1}^{n} y_{2j}\right)^2 + \cdots + \left(\sum_{j=1}^{n} y_{kj}\right)^2\right]$$

Here, $n = 5$ and $k = 4$. Therefore, we have

$$\frac{\sum_{i=1}^{4} y_{i.}^2}{5} = \frac{1}{5}(232^2 + 192^2 + 204^2 + 191^2) = 33757$$

Therefore,

$$\mathrm{SSTr} = 33757 - 33538.05 = 218.95$$

Block sum of squares is

$$\text{SSB} = \frac{\sum_{j=1}^{n} y_{.j}^2}{k} - \frac{y_{..}^2}{kn}$$

where

$$\frac{\sum_{j=1}^{n} y_{.j}^2}{k} = \frac{1}{k}(y_{.1}^2 + y_{.2}^2 + \cdots + y_{.n}^2) = \frac{1}{k}\left[\left(\sum_{i=1}^{k} y_{i1}\right)^2 + \left(\sum_{i=1}^{k} y_{i2}\right)^2 + \cdots + \left(\sum_{j=1}^{k} y_{in}\right)^2\right]$$

Here, $n = 5$ and $k = 4$. Therefore, we have

$$\frac{\sum_{j=1}^{5} y_{.j}^2}{4} = \frac{1}{4}(167^2 + 161^2 + 168^2 + 166^2 + 157^2) = 33559.75$$

Therefore,

$$\text{SSB} = 33559.75 - 33538.05 = 21.70$$

$$SSE = TSS - SSTr - SSB = 336.95 - 218.95 - 21.70 = 96.3$$

Degrees of Freedom (df)

For total: $nk - 1 = 20 - 1 = 19$

For treatment: $k - 1 = 4 - 1 = 3$

For block: $n - 1 = 5 - 1 = 4$

For Error: df (Total) – df (Treatment) – df (Block): $19 - 3 - 4 = (4 - 1)(5 - 1) = 12$

Now, the ANOVA table is constructed based on the above computations and the degrees of freedom as shown below.

Source of variation	*SS*	*df*	*MSS*	*F-ratio*
Between treatments	218.95	3	72.983	$F_{c1} = \frac{72.983}{8.025} = 9.094$
Between blocks	21.70	4	5.425	$F_{c2} = \frac{5.425}{8.025} = 0.676$
Error	96.30	12	8.025	
Total	32.78	19		

Inference

Since the calculated F-value, $F_{c1} = 9.094$, is greater than the tabulated value at 5% $[F_t = F(0.05; 3, 12) = 3.49]$ (refer to Tables E.2 in Appendix E), the hypothesis H_{01} is rejected. Also, since the calculated F-value $F_{c2} = 0.676$, is less than the tabulated value at 5% $[F_t = F(0.05; 4, 12) = 3.26]$ (refer to Tables E.2 in Appendix E), the hypothesis H_{02} is accepted. This implies that (i) there is a significant difference between the treatment effects and (ii) there is no significant difference between the block effects.